Business Statistics

Association between two variables



Some graphics and datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Association between two variables Overview

- How do a pair of variables relate to one another?
- Variance
- Covariance
- Correlation

Association between two variables

Association between two variables
To what degree do variables move together?

- Consider some basic questions about lifestyle and health:
  - Do taller people weigh more than shorter people?
  - Do people who exercise weigh less than those who do not?
  - Do people who exercise live longer?
- Data and statistics enable answering these types of questions
- Specifically, the two measures covariance and correlation provide a quantitative measure of the degree to which pairs of variables move together

Association between two variables
Data set: Lifestyle and physical characteristics

- How is exercise related to physical characteristics?
- Consider the below sample data of 5 people to investigate how age might be related to weight.
- The summary statistics below only provide quantitative information about each variable on its own, without regard for the other variables

Personal Body Characteristics				
Person	height (in)	weight (lbs)	age (yrs)	exercise (hrs/week)
A	62	175	52	1
B	65	155	28	6
C	68	195	60	2
D	71	165	35	5
E	74	185	45	3

Summary Stats				
Sample Statistic	height	weight	age	exer
Mean	68	175	44	3.4
Standard Deviation	4.7	15.8	12.8	2.1
Variance	22.5	250.0	164.5	4.3

Direction of a linear relationship

Covariance

Recall variance
Sample and population

- Variance measures how a single variable varies around its mean

$$s^2 = \frac{\sum_i (x_i - \overline{x})^2}{n-1} \qquad \sigma^2 = \frac{\sum_i (x_i - \mu)^2}{N}$$

- We define the numerator of the variance as the sum of the squares of all observations' deviations from the mean

$$\sum_i (x_i - \overline{x})^2 \qquad \sum_i (x_i - \mu)^2$$

- This can be expanded to

$$\sum_i (x_i - \overline{x})(x_i - \overline{x}) \qquad \sum_i (x_i - \mu)(x_i - \mu)$$

- Here we can rephrase the definition of the variance's numerator to be the sum of each observation's deviation **multiplied by itself**

Variance
Modified nomenclature

- So far the variance and standard deviation have been written as:

$$s^2 = \frac{\sum_i (x_i - \overline{x})^2}{n-1} \qquad \sigma^2 = \frac{\sum_i (x_i - \mu)^2}{N}$$

$$s = \sqrt{s^2} \qquad \sigma = \sqrt{\sigma^2}$$

- As the variance includes each observation's deviation multiplied by itself, s and σ can written with the variable names as subscripts

$$s_{xx} = \frac{\sum_i (x_i - \overline{x})(x_i - \overline{x})}{n-1} \qquad \sigma_{xx} = \frac{\sum_i (x_i - \mu)(x_i - \mu)}{N}$$

$$s_x = \sqrt{s_{xx}} \qquad \sigma_x = \sqrt{\sigma_{xx}}$$

Covariance
Co-movement of two variables about their means

- Variance measures how a single variable varies around its mean

$$s_{xx} = \frac{\sum_i (x_i - \overline{x})(x_i - \overline{x})}{n - 1}$$

- Covariance measures how two variables vary around their means together

$$s_{xy} = \frac{\sum_i (x_i - \overline{x})(y_i - \overline{y})}{n - 1}$$

- Covariance is calculated similar to variance, except it is for a pair of variable observations (x_i, y_i) instead of a single variable.

Covariance
The signs of the deviations

$$s_{xy} = \frac{\sum_i (x_i - \overline{x})(y_i - \overline{y})}{n - 1}$$

- A deviation's sign depends on the observation's value and the mean
  - An observation above the mean gives a positive deviation
  - An observation below the mean gives a negative deviation

$$x_i - \overline{x}$$

- Each covariance term's sign depends on the signs of the paired deviations

Sign of X deviation	X value relative to its mean	Sign of Y deviation	Y value relative to its mean	Covariance term	Movement of variables
+	Above	+	Above	+	Together
+	Above	-	Below	-	Opposite
-	Below	+	Above	-	Opposite
-	Below	-	Below	+	Together

- The final covariance sign depends on whether the positive or negative terms dominate in the sum

Association between two variables
Data set: Lifestyle and physical characteristics

- Recall the initial question.
- How is exercise related to physical characteristics?
- Consider how age might be related to weight for the sample data below for 5 people.
- The summary statistics provide quantitative information about each variable on its own, but not about relationships between the variables

Personal Body Characteristics				
Person	height (in)	weight (lbs)	age (yrs)	exercise (hrs/week)
A	62	175	52	1
B	65	155	28	6
C	68	195	60	2
D	71	165	35	5
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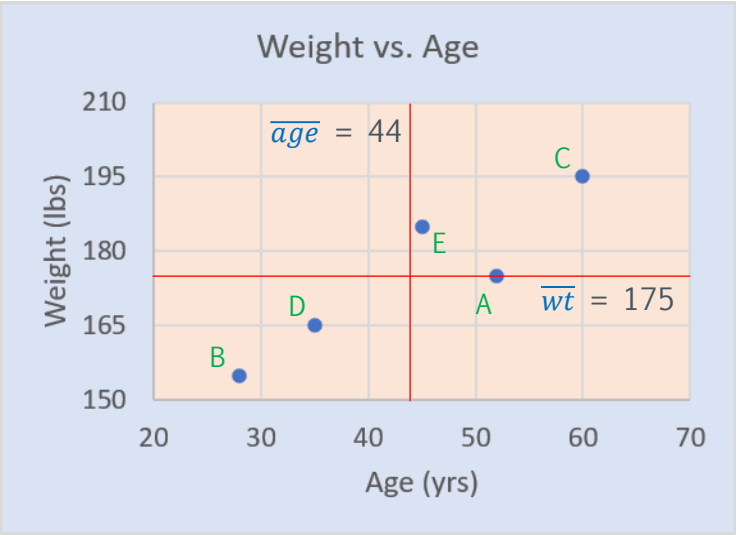
Summary Stats				
Sample Statistic	height	weight	age	exer
Mean	68	175	44	3.4
Standard Deviation	4.7	15.8	12.8	2.1
Variance	22.5	250.0	164.5	4.3

Recall visualizing two numerical variables Coordinate pairs and seeing if a relationship exists

- Take age $(\boldsymbol{x})$ and weight $(\boldsymbol{y})$ for each person as a pair of values
  - $(x_A, y_A) = (age_A, wt_A) = (52, 175)$ for person A
  - $(x_B, y_B) = (age_B, wt_B) = (28, 155)$ for person B
  - $\overline{x} = \overline{age} = 44$ and $\overline{y} = \overline{wt} = 175$

Personal Body Characteristics		
Person	age	weight
A	52	175
B	28	155
C	60	195
D	35	165
E	45	185
Mean	44	175

- Pairs of values are coordinates for a scatter plot
- The plot of these variables show that a linear relationship exists between weight and age
- However, it does not provide a **quantitative** measure of the direction or strength of that relationship



Covariance Calculation

$$s_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

- (age_i, wt_i) are the age and weight for person i

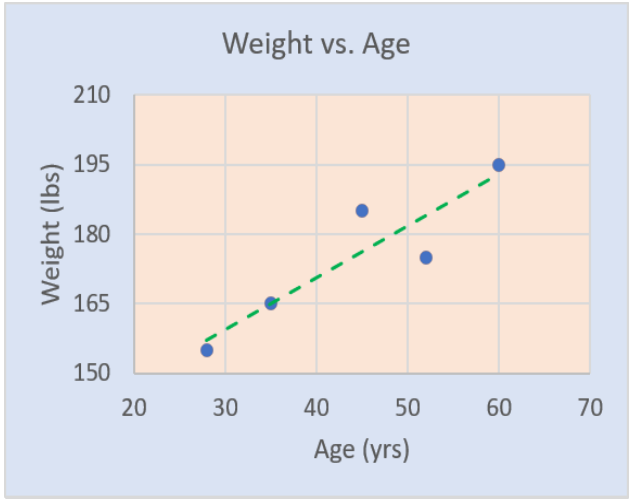
Personal Body Characteristics		
Person	age	weight
A	52	175
B	28	155
C	60	195
D	35	165
E	45	185
Mean	44	175

$$s_{Age,Wt} = \frac{\sum_i (age_i - \overline{age})(wt_i - \overline{wt})}{n - 1}$$

$$s_{Age,Wt} = \frac{(52 - 44)(175 - 175) + (28 - 44)(155 - 175) + \ \ldots + (45 - 44)(185 - 175)}{5 - 1}$$

$$s_{Age,Wt} = \frac{(8)(0) + (-16)(-20) + (16)(20) + (-9)(-10) + (1)(10)}{4}$$

$$s_{Age,Wt} = \frac{(0 + 320 + 320 + 90 + 10)}{4} = \frac{740}{4} = 185$$

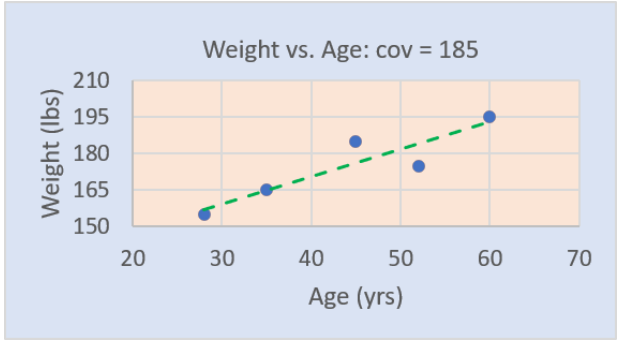


- Sample covariance between age and weight is 185

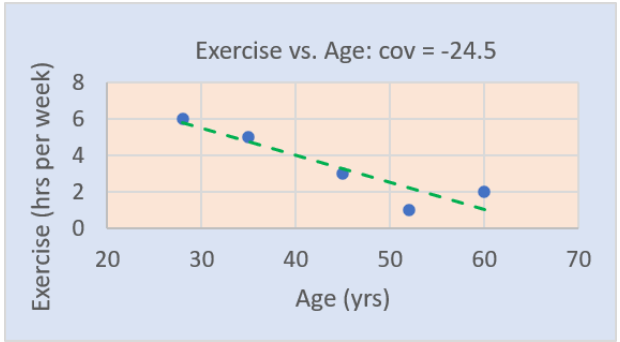
Covariance
Linear relationships

$$s_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

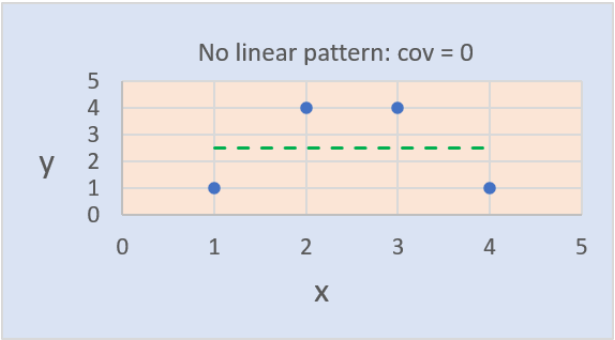
- Covariance > 0:
  - Positive linear relationship
  - On average, if x is above it's mean, y tends to be above it's mean, and vice versa



- Covariance < 0:
  - Negative linear relationship
  - On average, if x is above it's mean, y tends to be below it's mean, and vice versa



- Covariance = 0:
  - No linear relationship exists
  - There may be a relationship, but it is not linear



Covariance
The limitation of covariance

- The covariance sign indicates the direction of the relationship
- But the covariance magnitude depends on the products of different units of measurement
- Our data has weight in pounds, age in years, and exercise in hours
- What can be said when comparing these relationships:
  - Weight vs. Age (cov = 185 pound-years)
  - Exercise vs. Age (cov = -24.5 hours-years)
- Does weight have a stronger relationship with age than exercise?
- We cannot say, because 185 and -24.5 are completely different units

Strength of a linear relationship

Correlation

Correlation
Measures both direction and strength

- For the sample and population:
  - s_{xy} and σ_{xy} are covariances
  - s_x, s_y and σ_x, σ_y are standard deviations
- The sample and population correlation coefficients are:

$$r_{xy} = \frac{s_{xy}}{s_x s_y} \qquad \rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

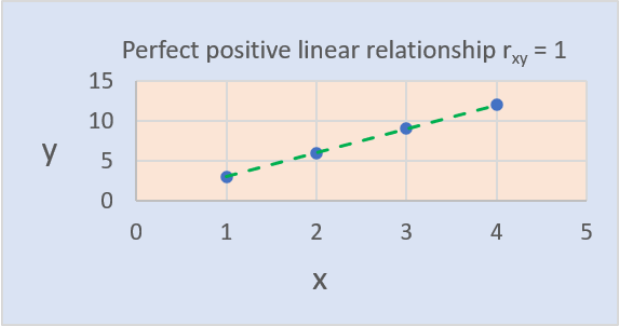
$$-1 \leq r_{xy} \leq 1 \qquad -1 \leq \rho_{xy} \leq 1$$

- Similar to covariance, the sign indicates the direction of the linear relationship
- The value indicates the strength:
  - $r_{xy} = 1$ indicates a perfect positive linear relationship between x and y
  - $r_{xy} = -1$ indicates a perfect negative linear relationship between x and y
  - $r_{xy} = 0$ indicates there is no linear relationship between x and y

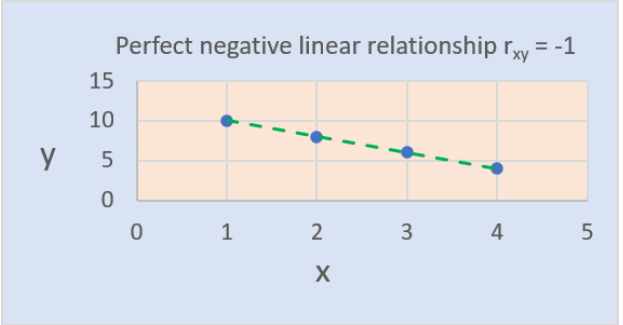
Correlation
Perfect and no linear relationships

$$r_{xy} = \frac{s_{xy}}{s_x s_y}$$

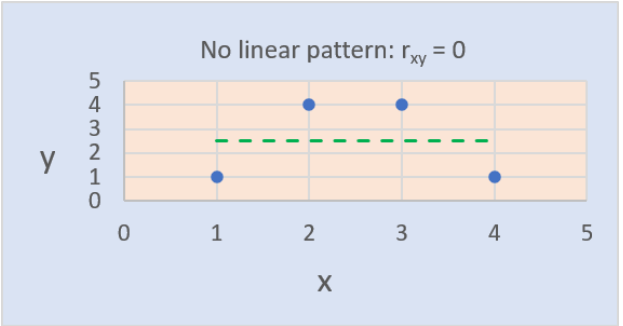
- $r_{xy} = 1$
  - Perfect positive linear relationship between x and y



- $r_{xy} = -1$
  - Perfect negative linear relationship between x and y



- $r_{xy} = 0$
  - No linear relationship exists between x and y



Correlation
Strength of the linear relationship

$$r_{xy} = \frac{s_{xy}}{s_x s_y}$$

- There are general guidelines for classifying the strength of the linear relationship:
  - Weak: $|r_{xy}| < 0.4$
  - Moderate: $0.4 \leq |r_{xy}| < 0.7$
  - Strong: $0.7 \leq |r_{xy}|$

- These are general guidelines as they depend on the context, sample size, and subject of the analysis